

Persistent Homology Reveals Meditation Expertise-Related Regularity in Auditory Oddball EEG Dynamics

Chandan Kumar Srivastava^{1,*}, Tapas Rath², Jyotiranjana Beuria³, Rashmi Gupta¹

¹Department of Humanities and Social Sciences, Indian Institute of Technology Bombay, Mumbai, India

²Department of Cognitive Science, Indian Institute of Technology Kanpur, Kanpur, India

³Indian Knowledge System and Mental Health Center, Indian Institute of Technology Mandi, Mandi, India

*Correspondence: srivastava.chandan@iitb.ac.in

Abstract

Mismatch negativity (MMN) is widely used to probe automatic auditory change detection, but conventional event-related potential amplitudes may miss differences in the organization of auditory responses that give rise to MMN waveforms. We re-analysed Fz-channel EEG from an intensity-oddball paradigm comparing 27 experienced mantra meditators and 27 novice practitioners. We then applied topological analysis to the standard and deviant auditory responses used to derive the MMN difference waveforms. Specifically, single-trial waveforms were embedded using delay coordinates, and Vietoris–Rips persistent homology was quantified in dimensions H_0 and H_1 . Matched MMN and entropy comparisons used two-sided Welch-studentized participant-label permutation tests with 100,000 reallocations. Conventional MMN amplitudes showed no reliable expertise effects ($p = 0.286$ and 0.223). Experts showed higher mean raw H_0 entropy and lower across-trial entropy variability in standards and both deviant responses at Fz; all six focal comparisons had the hypothesized directions (permutation $p = 0.00001$ to 0.00668). Mean raw H_1 estimates were also higher in experts, but did not establish a group difference. Cz analyses are reported in the supplement. These findings support persistent homology as a complementary description of auditory response organization. Greater alertness in experienced meditators is one possible interpretation, but alertness was not measured directly.

Keywords: auditory oddball; electroencephalography; mismatch negativity; meditation expertise; persistent homology; topological data analysis

1 Introduction

Mismatch negativity (MMN) is widely recognized as a neurophysiological marker of automatic auditory change detection, reflecting the brain’s involuntary response to violations of auditory regularities (Näätänen et al., 2007). Studies investigating the effects of meditation practice on change detection have examined modulation of the MMN; however, the evidence remains mixed (Fucci et al., 2022; Biedermann et al., 2016; Srinivasan and Baijal, 2007; Fucci et al., 2018).

In the present study, we revisit the intensity-oddball dataset from the original investigation (Srivastava et al., 2025), which was conducted in the context of these mixed findings. The original analysis found no statistically significant differences in conventional MMN amplitude between experienced mantra meditators and novice practitioners. This result is scientifically informative, as discussed in the original study, but it leaves open the question of whether meditation expertise is reflected in the organization of the underlying auditory responses. MMN amplitude is a group comparison of deviant-minus-standard averaged waveforms; it does not describe every property of the separate standard and deviant trials.

Complementary EEG measures may therefore address questions that are distinct from the amplitude of a difference ERP. For example, Lu and Rodriguez-Larios used nonlinear EEG measures to study mind wandering during breath-focused meditation (Lu and Rodriguez-Larios, 2022). Their task and outcome differ from the present auditory-oddball expertise comparison, but the study illustrates the value of specifying which property of the EEG signal a measure is intended to summarize. Here, we use delay-coordinate embedding and persistent homology to describe the geometry of condition-specific EEG trajectories. These summaries are mathematically nonlinear functions of the waveform.

Topological data analysis (TDA) offers one way to test this possibility. Neural activity exhibits complex temporal dynamics that may not be fully captured by conventional measures of response magnitude. By transforming a waveform into a delay-coordinate point cloud and summarizing its persistent homology, TDA can capture geometric organization that is not reducible to mean voltage within a fixed component window (Takens, 1981; Edelsbrunner et al., 2002; Ghrist, 2008; Perea and Harer, 2015). In EEG, this is especially attractive when the hypothesis concerns trial-level stability or dynamical regularity rather than a simple increase or decrease in ERP amplitude. Raw H_0 persistence entropy summarizes the distribution of connected-component lifetimes in a trial’s persistence diagram; its within-participant trial SD summarizes the variability of that feature across trials. Neither summary directly measures alertness.

Here, we ask whether persistent-homology summaries of auditory-evoked EEG responses at frontocentral electrodes reveal differences between experienced and novice practitioners in the same dataset where conventional ERP amplitudes did not. Before inspecting this cohort’s topological features, we specified the directional hypothesis that experts would have higher mean raw H_0 entropy and lower within-participant trial-to-trial variability at Fz for standards and both deviant types. We report matched Fz MMN comparisons alongside these six focal entropy comparisons, examine mean raw H_1 entropy as a secondary outcome, and place Cz electrode sensitivity analyses in the supplement.

2 Materials and methods

2.1 Participants

The present analysis reused the curated dataset reported in the original study (Srivastava et al., 2025) and did not recruit new participants. The source study was approved by the Ethical Review Board of the Indian Institute of Technology Bombay, and informed consent was obtained from all participants. The cohort included 54 participants (27 experienced mantra meditators and 27 novice practitioners). Experts had at least 10 years of practice (approximately 120 minutes/day), whereas novices had 1 year or less of practice (approximately 10–20 minutes/day). Group ages were comparable (experts: mean = 29.9 years, SD = 3.8; novices: mean = 28.1 years, SD = 6.4). The sample was predominantly male (25 males and 2 females in each group, as reported in the source dataset), with normal hearing and no reported history of psychiatric or neurological disorders.

2.2 Experiment Procedure and Task Apparatus

Participants were exposed to an intensity oddball paradigm consisting of two blocks, each containing 500 auditory stimuli. In each block, 80% of the stimuli were standard tones presented at 80 dB, while the remaining 20% comprised equal proportions of louder (90 dB) and quieter (70 dB) deviants (10% each). Consequently, each block contained 50 trials of the 90 dB deviant (D90), 50 trials of the 70 dB deviant (D70), and 400 trials of the 80 dB standard (S80). The interstimulus interval varied randomly between 450 and 550 ms, with values drawn from a uniform distribution. Participants engaged in a distractor task by fixating on a cross displayed

on the screen and counting its colour changes, which randomly alternated between black and gray, while passively listening to the auditory tones and being instructed to ignore them.

EEG signals were recorded while participants were exposed to the auditory stimuli and engaged in the distractor task. Signals were acquired at a sampling frequency of 250 Hz using a 32-channel EEG recording system, in accordance with the protocols detailed in the source study (Srivastava et al., 2025).

2.3 Data Analysis

Preprocessed EEG set files from the original study were used without altering its artifact-handling pipeline. The source preprocessing steps included filtering, artifact handling, referencing, and epoch construction, as reported in the source study (Srivastava et al., 2025). We first re-analysed the conventional MMN obtained from these data using permutation-based statistics to assess the robustness of the original findings. We then applied topological data analysis as the central analysis of the present study. Fz was the focal electrode for the matched MMN and topology comparisons; Cz was examined as an electrode sensitivity analysis in the supplement.

2.3.1 Shared waveform preparation

We re-analysed the retained EEG epochs from the source study’s 0.5-Hz high-pass branch. For each retained trial, voltage was converted to microvolts and its mean over the full saved epoch was subtracted. A zero-phase fourth-order Butterworth 20-Hz low-pass filter was then applied to the complete individual trial before cropping, ERP averaging, difference-wave construction, or topological feature extraction. The resulting trials were restricted to $-100 \leq t < 450$ ms for the analyses below; at 250 Hz the last included sample was 448 ms. Thus, the conventional and topological analyses used the same prepared trial branch.

2.3.2 Conventional ERP Data

To obtain standard and deviant event-related potentials (ERPs), all retained trials corresponding to S80, D70, and D90 were averaged separately within each participant at Fz. Conventional MMN difference waveforms were calculated by subtracting each participant’s S80 ERP from the D90 and D70 ERPs; individual trials were not paired across conditions. Each difference waveform was then baseline corrected by subtracting its mean voltage from -100 to < 0 ms. Conventional MMN amplitudes were calculated as the mean amplitudes of these baseline-corrected D90–S80 and D70–S80 waveforms within the $125 \leq t < 225$ ms measurement window (25 samples, 128–224 ms). Thereafter, permutation-based statistics were applied to the single MMN score per participant and contrast. Figure 2 displays the difference waves and participant-level scores.

2.4 Topological Data

Topological data analysis was applied to single-trial standard and deviant waveforms (S80, D90, and D70) obtained from the same preprocessed EEG files at Fz and Cz. The analysis window included the pre-stimulus baseline and extended from -100 to < 450 ms. Each cropped trial was represented using time-delay embedding with an embedding dimension of 5, a delay of 3 samples, and a unit stride. The delay corresponded to 12 ms, and the embedding span was 48 ms. Vietoris–Rips persistent homology was computed using Euclidean distance in dimensions H_0 and H_1 , with $\mathbb{Z}_2$ coefficients, reduced H_0 , edge collapse, and a filtration cap of $30 \mu\text{V}$, as implemented in giotto-tda. Raw persistence entropy was calculated for each trial’s persistence diagram. For each participant and stimulus condition, mean raw H_0 entropy was calculated across trials; trial-to-trial raw H_0 entropy variability was calculated as the within-participant sample standard deviation. Mean raw H_1 entropy was examined as a secondary outcome. These

Topology of auditory oddball EEG

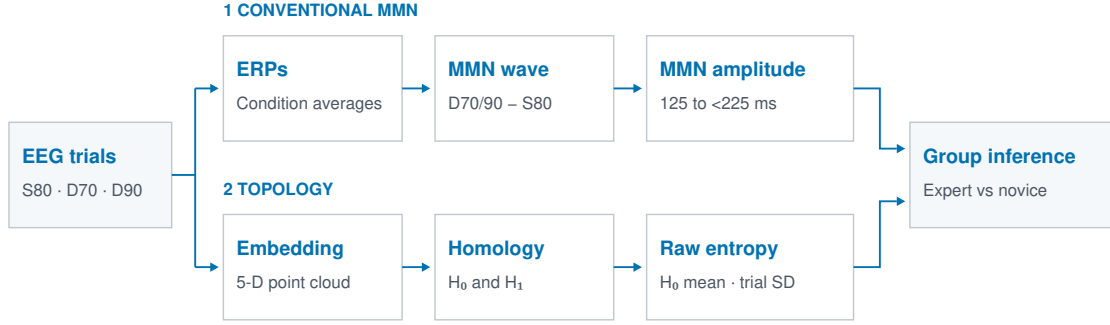


Figure 1: Schematic workflow for condition-specific auditory EEG topology and the matched conventional MMN comparison. Fz raw H_0 entropy mean and within-participant trial SD were the focal outcomes for S80, D70, and D90; raw H_1 entropy was a secondary outcome. The MMN and entropy group comparisons used the same participant-level, two-sided Welch-studentized permutation procedure with 100,000 reallocations. Cz analyses are reported in the supplement

measures were computed from the separate condition-specific responses, not from the MMN difference waveforms.

2.5 Statistical Analysis

Differences between expert and novice groups in MMN mean amplitude, mean raw H_0 persistence entropy, and trial-to-trial raw H_0 entropy variability were quantified using Hedges’ g , calculated as the mean value for experts minus the mean value for novices and corrected for small-sample bias. Positive values indicate higher values in experts, whereas negative values indicate lower values in experts. Mean raw H_1 entropy was treated as a secondary comparison.

Permutation-based statistics were applied to test for group differences in conventional ERP amplitude and topological metrics. The participant was the independent unit of inference for every comparison ($n_E = n_N = 27$). The signed Welch-studentized statistic was

$$T = \frac{\bar{x}_E - \bar{x}_N}{\sqrt{s_E^2/n_E + s_N^2/n_N}}, \quad (1)$$

where s_E^2 and s_N^2 are group sample variances. Two-sided permutation tests compared $|T|$ with its empirical distribution from 100,000 random reallocations of participant labels, preserving 27 participants per group and recomputing the denominator for each reallocation. With b permuted statistics at least as extreme as observed, the Monte Carlo p -value was $(b + 1)/100001$. We estimated 95% confidence intervals using 10,000 stratified bootstrap resamples, preserving the expert and novice group sizes. Bootstrap intervals quantify effect-size uncertainty and are distinct from permutation tests. Pointwise waveform bands are descriptive, not time-resolved tests.

The same test statistic, permutation count, two-sided tail rule, and bootstrap procedure were used for the two conventional MMN contrasts and the six focal Fz entropy comparisons. We report their endpoint-level permutation p -values directly, without a multiplicity adjustment. Although the focal entropy comparisons were hypothesis-led, their p -values should be read with the number of comparisons in mind. Figure 1 summarizes the analysis paths.

Table 1: Raw H_0 persistence entropy at Fz. Values are participant-level means across trials, reported as group mean (SD); Hedges’ g is expert minus novice with 95% stratified percentile-bootstrap confidence intervals. Permutation p -values use 100,000 two-sided Welch-studentized participant-label reallocations

Condition	Experts	Novices	Hedges’ g [95% CI]	p
S80	6.8454 (0.0048)	6.8308 (0.0248)	0.806 [0.539, 1.132]	0.00024
D70	6.8453 (0.0076)	6.8313 (0.0255)	0.732 [0.365, 1.083]	0.00421
D90	6.8462 (0.0060)	6.8310 (0.0218)	0.935 [0.653, 1.306]	0.00007

2.6 AI-assisted manuscript preparation

During the preparation of this work, the authors used ChatGPT and Grammarly to assist with language editing and readability. The authors reviewed and edited the content as needed and take full responsibility for the content of the article.

3 Results

3.1 Conventional MMN Amplitude Results

Conventional MMN mean amplitudes were first re-analysed to assess the robustness of the original findings using the same participant-level permutation procedure applied to the topological measures. For D90–S80, the mean amplitude was $-1.775 \mu\text{V}$ (SD 1.140) in experts and $-2.139 \mu\text{V}$ (SD 1.327) in novices. The expert-minus-novice difference was $0.364 \mu\text{V}$ (95% bootstrap CI $[-0.276, 1.006]$), with Hedges’ $g = 0.290$ (95% CI $[-0.235, 0.830]$) and a two-sided permutation $p = 0.286$. For D70–S80, the corresponding group means were $-0.375 \mu\text{V}$ (SD 1.057) and $-0.711 \mu\text{V}$ (SD 0.951); the difference was $0.337 \mu\text{V}$ (95% CI $[-0.187, 0.866]$), $g = 0.330$ (95% CI $[-0.189, 0.942]$), and $p = 0.223$. These comparisons did not provide clear evidence of an expertise difference in conventional MMN amplitude (Fig. 2). Their confidence intervals include effects in both directions and do not establish equivalence between groups.

3.2 Topological Feature Results

3.2.1 H_0 persistence entropy

Experts showed higher mean raw H_0 persistence entropy than novices in the standard and deviant waveforms that give rise to the MMN waveform when topological data analysis was applied to trial-level EEG responses. At Fz, this difference was observed for S80 ($g = 0.806$, $p = 0.00024$), D70 ($g = 0.732$, $p = 0.00421$), and D90 ($g = 0.935$, $p = 0.00007$). The positive direction was consistent across the three condition-specific trial sets (Table 1; Fig. 3A).

3.2.2 H_0 persistence entropy variability

This higher mean raw H_0 persistence entropy was accompanied by lower trial-to-trial entropy variability in experts compared to novices. The within-participant standard deviation was lower in experts for S80 ($g = -1.005$, $p = 0.00001$), D70 ($g = -0.720$, $p = 0.00275$), and D90 ($g = -0.727$, $p = 0.00668$). All six focal H_0 comparisons were in the hypothesized directions (Table 2; Fig. 3B). The corresponding Cz comparisons are reported only in Supplementary Table S1 and Supplementary Fig. S1.

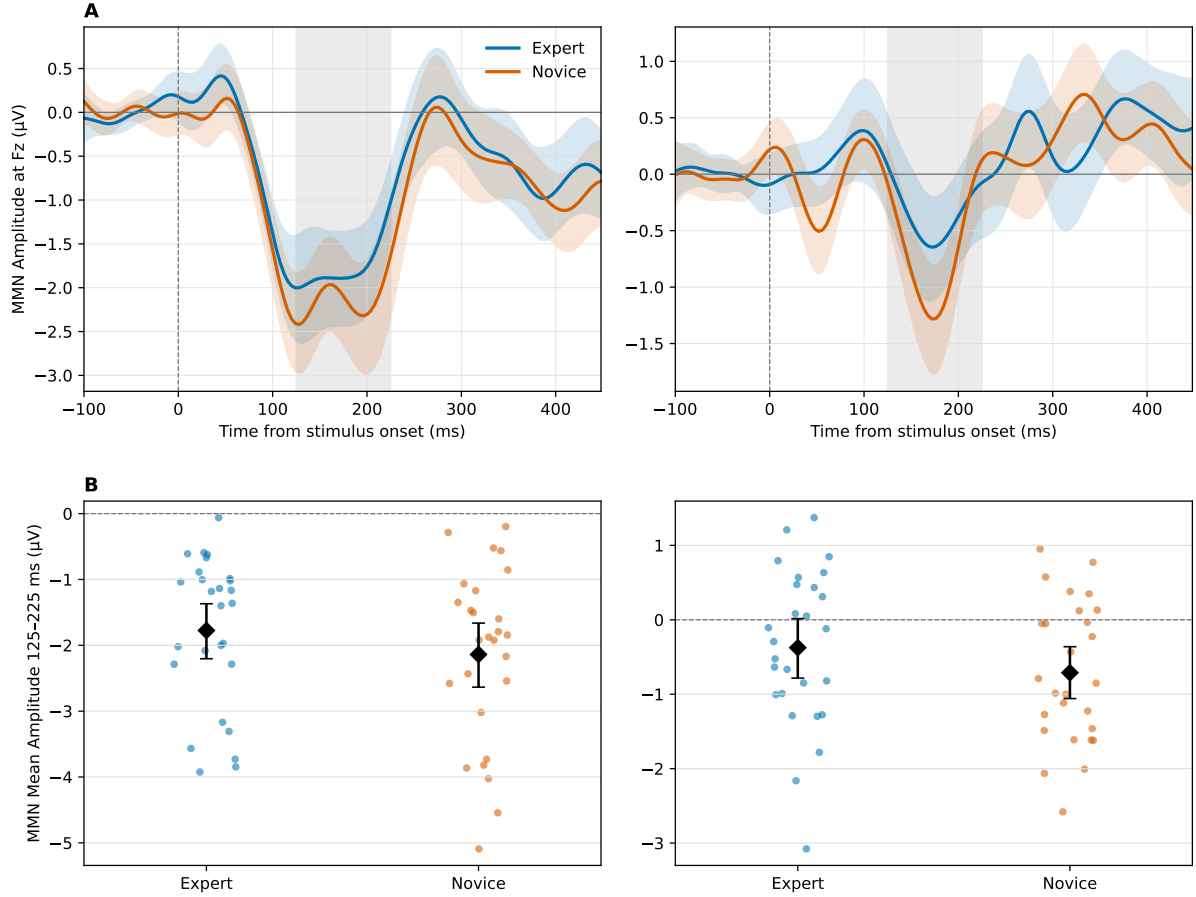


Figure 2: Matched conventional Fz MMN responses. (A) Group-mean, baseline-corrected D90–S80 (left) and D70–S80 (right) difference ERPs; bands show descriptive pointwise 95% bootstrap confidence intervals for the group means. Gray shading marks the 125–225 ms measurement window. (B) Participant-level window-mean amplitudes for the same contrasts; diamonds and bars show group means and 95% bootstrap confidence intervals. Each group has 27 participants. The two-sided 100,000-reallocation permutation p -values are 0.286 and 0.223, respectively. Waveform bands are not time-resolved significance tests

3.2.3 H_1 persistence entropy

At Fz, mean raw H_1 persistence entropy was numerically higher in experts than novices for S80, D70, and D90 (Hedges' $g = 0.54, 0.50$, and 0.47 , respectively; two-sided permutation $p = 0.052, 0.070$, and 0.084 ; Supplementary Fig. S2). The estimates share a positive direction, but these tests do not establish an expert–novice difference in H_1 entropy. The corresponding Cz results are in Supplementary Table S2.

4 Discussion

The present study aimed to investigate whether expertise in mantra meditation is associated with differences in the topological properties of single-trial auditory-evoked EEG responses, despite the absence of clear group evidence in conventional MMN amplitude. Using topological data analysis, we examined the persistent-homology features of standard and deviant auditory responses in experienced mantra meditators and novice practitioners. Specifically, we investigated whether differences in mean raw H_0 persistence entropy and its trial-to-trial variability, with mean raw H_1 entropy as a secondary outcome, could provide complementary insights into the

Table 2: Trial-to-trial raw H_0 persistence entropy variability at Fz. Values are within-participant sample standard deviations across trials, reported as group mean (SD); Hedges’ g is expert minus novice with 95% stratified percentile-bootstrap confidence intervals. Permutation p -values use 100,000 two-sided Welch-studentized participant-label reallocations

Condition	Experts	Novices	Hedges’ g [95% CI]	p
S80	0.0566 (0.0041)	0.0706 (0.0190)	−1.005 [−1.483, −0.745]	0.00001
D70	0.0551 (0.0069)	0.0668 (0.0215)	−0.720 [−1.132, −0.388]	0.00275
D90	0.0578 (0.0073)	0.0666 (0.0151)	−0.727 [−1.168, −0.278]	0.00668

organization of auditory-evoked neural responses that may not be captured by conventional ERP measures.

Results showed that experts had higher raw H_0 persistence entropy than novices in the standard and deviant waveforms that give rise to the MMN waveform when topological data analysis was applied to trial-level EEG responses. This result was accompanied by lower trial-to-trial raw H_0 persistence entropy variability in experts compared to novices. The raw H_1 entropy estimates were also higher in experts for standards and both deviants, but their two-sided permutation tests did not establish an expert–novice difference ($p = 0.052, 0.070$, and 0.084).

One possible interpretation is that higher H_0 persistence entropy may reflect greater alertness in experts during the processing of standard and oddball responses, whereas lower trial-to-trial entropy variability may indicate that such a state is more consistent across trials. Previous work has related other EEG complexity measures to human alertness and to meditation-related states (Hadra et al., 2022; Lu and Rodriguez-Larios, 2022; Walter and Hinterberger, 2022). Those studies used different tasks and measures, however, and alertness was not independently recorded here. The present entropy results therefore motivate an alertness hypothesis; they do not demonstrate that the experts were more alert or establish raw H_0 persistence entropy as an alertness marker. Lower trial SD also could reflect measurement noise, retained-trial composition, filtering, or other unmeasured differences.

Thus, the main finding is that persistent-homology features of single-trial EEG responses from standard and oddball stimuli in an intensity oddball paradigm revealed expertise-related differences where conventional MMN amplitudes did not reliably separate experienced mantra meditators from novices. The existence of topology-related differences in the waveforms that contribute to MMN suggests that expertise-related differences may be expressed in the geometry and variability of the raw auditory responses that contribute to ERP and MMN estimates. The present findings can be used as preliminary evidence to motivate hypotheses for topology-based measures of meditation expertise and, more generally, for assessing EEG markers of individual differences. These measures characterize separate condition-specific responses; they are not topological features of the MMN difference wave itself.

Several limits qualify this interpretation. The conventional MMN confidence intervals do not establish equal amplitudes, and the present analysis does not establish that topology is more sensitive than MMN. Sound level and adaptation may influence condition-specific responses, so the entropy results cannot isolate deviance detection or identify a neural mechanism. The visual distractor and the novices’ prior practice may also influence responses, but neither influence was manipulated in this analysis.

The cross-sectional expert–novice comparison does not identify a causal effect of meditation training. The sample is modest, predominantly male, and drawn from one source study. Cz uses the same participants and appears only in the supplement. Future work should prospectively document the entropy contrasts and analysis plan, assess signal quality and trial counts, recruit a more diverse cohort, and test whether these associations replicate out of sample while also measuring alertness directly.

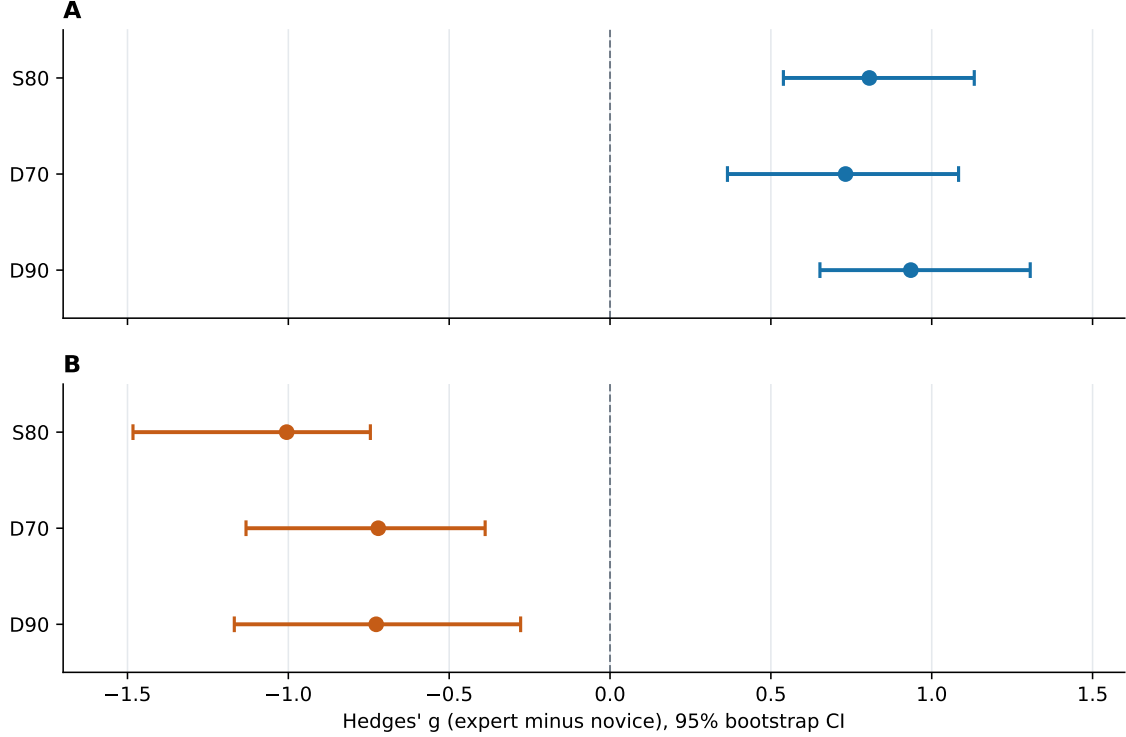


Figure 3: Expert–novice Hedges' g for raw H_0 persistence entropy at Fz. (A) Each participant's mean entropy across trials and (B) trial-to-trial entropy variability for S80, D70, and D90. Points represent expert-minus-novice effects; horizontal bars are 95% bootstrap confidence intervals, and the dashed vertical line denotes no group difference. Positive values indicate higher entropy in experts; negative values in panel B indicate lower variability in experts. Two-sided permutation p -values are in Tables 1 and 2

5 Conclusion

Persistent homology revealed consistent expertise-related differences in Fz auditory oddball EEG responses. Experts showed higher trial-level raw H_0 entropy and lower raw H_0 entropy variability than novices, while matched conventional MMN amplitudes did not reliably distinguish the groups. These results motivate the use of topology-based analyses as complementary tools for studying neural response organization in auditory cognitive neuroscience, with the alertness interpretation to be tested directly in future work.

Statements and Declarations

Ethics approval and consent to participate. This secondary analysis used data from the source study, which reported ethics approval from the Ethical Review Board of the Indian Institute of Technology Bombay and informed consent from all participants (Srivastava et al., 2025). No new participants were recruited.

Consent for publication. No identifiable participant material is included.

Competing interests. The authors declare no competing interests.

Funding. No specific funding was received for this work.

Author contributions. Chandan Kumar Srivastava: conceptualization, methodology, investigation, data curation, formal analysis, original draft, review and editing. Tapas Rath: formal analysis, methodology, original draft, review and editing. Jyotiranjana Beuria: formal analysis, methodology, review and editing. Rashmi Gupta: conceptualization, funding acquisition,

resources, supervision, review and editing.

Data and code availability. Derived participant-level analysis tables and group-analysis scripts are available from the corresponding authors upon reasonable request. Access to underlying EEG data is subject to the source study’s ethics and participant-consent constraints.

References

- Biedermann B, De Lissa P, Mahajan Y, Polito V, Badcock N, Connors MH, Quinto L, Larsen L, McArthur G (2016) Meditation and auditory attention: an ERP study of meditators and non-meditators. *Int J Psychophysiol* 109:63–70. <https://doi.org/10.1016/j.ijpsycho.2016.09.016>
- Edelsbrunner H, Letscher D, Zomorodian A (2002) Topological persistence and simplification. *Discrete Comput Geom* 28:511–533. <https://doi.org/10.1007/s00454-002-2885-2>
- Fucci E, Abdoun O, Caclin A, Francis A, Dunne JD, Ricard M, Davidson RJ, Lutz A (2018) Differential effects of non-dual and focused attention meditations on the formation of automatic perceptual habits in expert practitioners. *Neuropsychologia* 119:92–100. <https://doi.org/10.1016/j.neuropsychologia.2018.07.025>
- Fucci E, Pouban-Couzardot A, Abdoun O, Lutz A (2022) No effect of focused attention and open monitoring meditation on EEG auditory mismatch negativity in expert and novice practitioners. *Int J Psychophysiol* 176:62–72. <https://doi.org/10.1016/j.ijpsycho.2022.03.010>
- Ghrist R (2008) Barcodes: the persistent topology of data. *Bull Am Math Soc* 45:61–75. <https://doi.org/10.1090/S0273-0979-07-01191-3>
- Hadra M, Omidvarnia A, Mesbah M (2022) Temporal complexity of EEG encodes human alertness. *Physiol Meas* 43:095002. <https://doi.org/10.1088/1361-6579/ac8f80>
- Lu Y, Rodriguez-Larios J (2022) Nonlinear EEG signatures of mind wandering during breath focus meditation. *Curr Res Neurobiol* 3:100056. <https://doi.org/10.1016/j.crneur.2022.100056>
- Näätänen R, Paavilainen P, Rinne T, Alho K (2007) The mismatch negativity (MMN) in basic research of central auditory processing: a review. *Clin Neurophysiol* 118:2544–2590. <https://doi.org/10.1016/j.clinph.2007.04.026>
- Perea JA, Harer J (2015) Sliding windows and persistence: an application of topological methods to signal analysis. *Found Comput Math* 15:799–838. <https://doi.org/10.1007/s10208-014-9206-z>
- Srinivasan N, Baijal S (2007) Concentrative meditation enhances preattentive processing: a mismatch negativity study. *NeuroReport* 18:1709–1712. <https://doi.org/10.1097/WNR.0b013e3282f0d2d8>
- Srivastava C, O’Reilly JA, Gupta R (2025) Investigating the effects of focused attention (mantra) meditation on mismatch negativity: insights into sensory and cognitive processing using an intensity oddball paradigm. *Neuroscience* 571:62–73. <https://doi.org/10.1016/j.neuroscience.2025.02.017>
- Takens F (1981) Detecting strange attractors in turbulence. In: *Dynamical Systems and Turbulence*, Warwick 1980. Springer, pp 366–381. <https://doi.org/10.1007/BFb0091924>
- Walter N, Hinterberger T (2022) Determining states of consciousness in the electroencephalogram based on spectral, complexity, and criticality features. *Neurosci Conscious* 2022:niac008. <https://doi.org/10.1093/nc/niac008>

Supplementary material

Persistent Homology Reveals Meditation Expertise-Related Regularity in Auditory Oddball EEG Dynamics

Chandan Kumar Srivastava, Tapas Rath, Jyotiranjana Beuria, and Rashmi Gupta

Cz electrode sensitivity analyses

Cz comparisons used the same 54 participants as the focal Fz tests. The directions of the raw- H_0 mean and trial-SD effects were consistent with Fz (Table S1; Fig. S1). The secondary raw- H_1 comparisons are shown in Table S2.

Table S1: Cz sensitivity comparisons for raw H_0 persistence-entropy summaries. Hedges' g is expert minus novice with 95% stratified percentile-bootstrap confidence intervals. As for conventional MMN and the focal Fz analyses, p -values use 100,000 two-sided Welch-studentized participant-label permutations

Summary	Condition	g [95% CI]	Permutation p
Mean	S80	0.696 [0.354, 1.037]	0.00586
	D70	0.705 [0.331, 1.070]	0.00800
	D90	0.696 [0.327, 1.059]	0.00616
Trial SD	S80	-0.942 [-1.341, -0.647]	0.00006
	D70	-0.680 [-1.057, -0.346]	0.00529
	D90	-0.746 [-1.191, -0.311]	0.00648

Table S2: Secondary Cz raw- H_1 entropy comparisons. Hedges' g is expert minus novice; p -values use two-sided Welch-studentized participant-label permutation tests with 100,000 reallocations

Summary	Condition	Hedges' g	Permutation p
Mean raw H_1 entropy	S80	0.405	0.13645
	D70	0.416	0.12573
	D90	0.369	0.17415

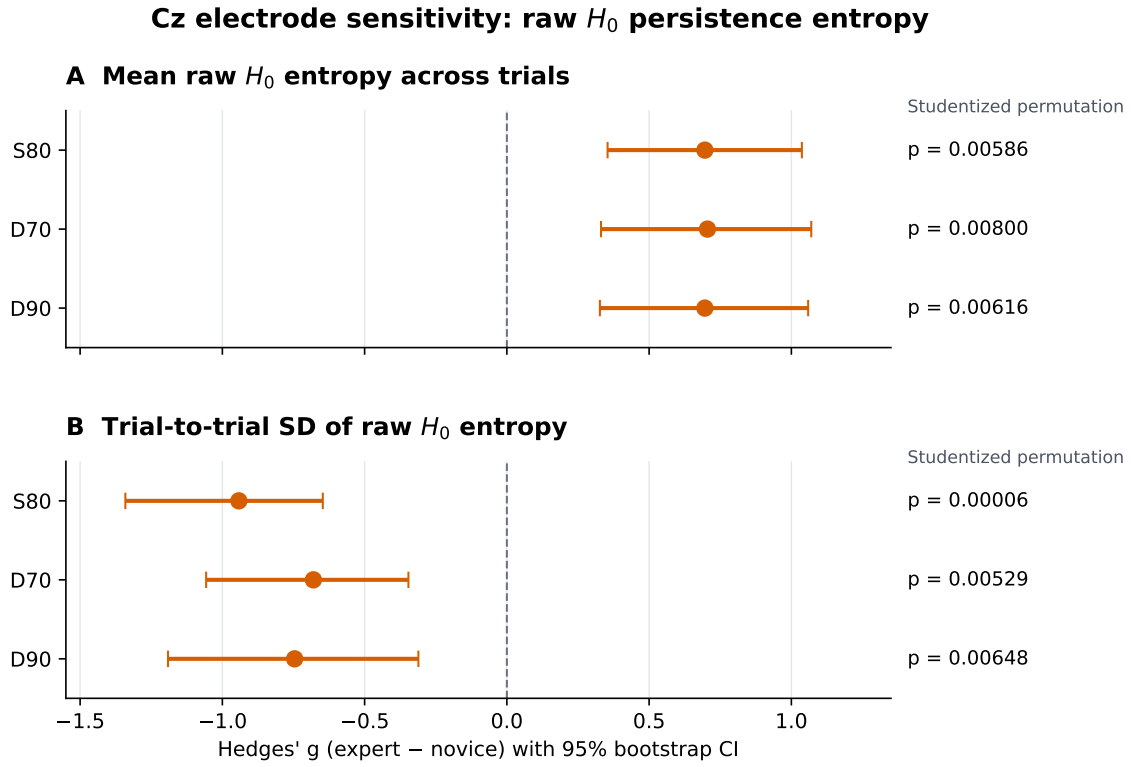


Figure S1: Raw- H_0 mean and trial-SD expert-minus-novice effects at Fz and Cz. Points are Hedges' g and bars are 95% stratified percentile-bootstrap intervals. Right-hand p -values use two-sided Welch-studentized participant-label permutations with 100,000 reallocations

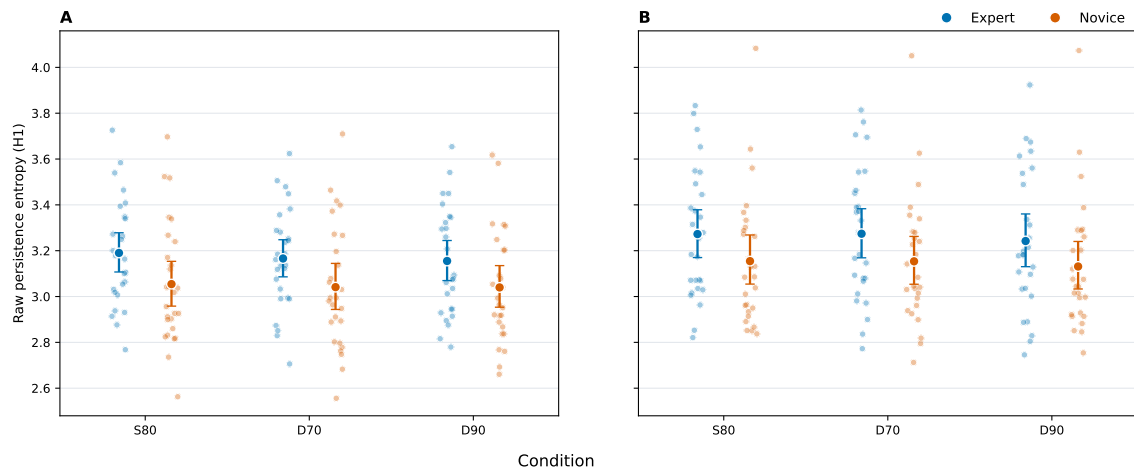


Figure S2: Participant-level raw H_1 persistence entropy in S80, D70, and D90 at Fz (A) and Cz (B). Small points represent individual participants; large points and bars are group means and descriptive 95% percentile-bootstrap intervals. Each group includes 27 participants. The Fz permutation results are reported in the main text; the Cz results are exploratory sensitivity analyses